Q32 — Discussion

Quantifying information flow with entropy

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1. **Entropy vs. correctness.** A model answers a factual question with very low next-token entropy but is wrong. A second model answers with high entropy and is right. What does each case tell you, and why is "confident" not the same as "calibrated"? (Connect to Q11/Q12.)

2. **The irreducible term.** Cross-entropy decomposes into $H(p_{\text{data}}) + D_{\text{KL}}(p_{\text{data}} \| p_\theta)$. If a lab reports a new model with lower perplexity on the same corpus, which term moved — and could a reported perplexity gain ever be an artefact of the *tokenizer* rather than the model? How would you control for that?

3. **The information-plane debate.** Shwartz-Ziv & Tishby saw a "compression phase"; Saxe et al. showed it depends on the nonlinearity and the binning. If a classmate shows you an information-plane plot of a network, what three questions do you ask before believing it?

4. **Attention entropy as interpretation.** A given head has near-zero entropy on syntax tokens but near-maximal entropy on content words. Is this head "doing more" on syntax? What alternative explanations (induction behaviour, positional averaging, a "no-op"/attention-sink head) could produce the same entropy signature, and how would you distinguish them?

5. **Estimate vs. intervene.** You want to know whether information about token i reaches the prediction at position j. One student proposes estimating $I(h_i; \hat{y}_j)$ with MINE; another proposes activation patching. Under what conditions does each answer a *different* question, and which one answers the *causal* question you probably care about?